

Astrophysical cosmic rays

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PROBLEM SET

1. Energy threshold for the proton GZK

Show that in a frame where the CMB photon has energy E_γ , the proton energy thresholds for the photopion reactions

$$p_{\text{UHECR}} + \gamma_{\text{CMB}} \longrightarrow p + \pi^0 \quad \text{or} \quad p_{\text{UHECR}} + \gamma_{\text{CMB}} \longrightarrow n + \pi^+ \quad (1)$$

are given by:

$$E_p = \frac{[(m_f c^2 + m_\pi c^2)^2 - m_i^2 c^4]^2 + 4m_i^2 c^4 E_\gamma^2}{4E_\gamma [(m_f c^2 + m_\pi c^2)^2 - m_i^2 c^4]} \simeq \frac{(m_f c^2 + m_\pi c^2)^2 - m_i^2 c^4}{4E_\gamma}, \quad (2)$$

where m_i and m_f are the rest masses of the barions in the initial and final states, respectively, and m_π the mass of the appropriate pion in the final state. Justify the approximation used in the last step. Calculate the proton energy at this threshold for a photon with an energy equal to the average energy of the CMB Planck distribution today.

2. UHECRs adiabatic energy losses

Consider a collection of UHECR propagating in an expanding universe, and show that the rate of change of its energy due to the expansion satisfies:

$$\frac{1}{E} \left(\frac{dE}{dt} \right)_{\text{ad}} = -H(t), \quad (3)$$

where $H(t)$ is the Hubble parameter. This energy loss, due solely to the universe expansion, is called adiabatic energy loss. Hint: write down the continuity equation for the energy-momentum tensor

$$T^\mu{}_{\nu;\mu} = T^\mu{}_{\nu,\mu} + \Gamma^\mu{}_{\alpha\mu} T^\alpha{}_\nu - \Gamma^\alpha{}_{\nu\mu} T^\mu{}_\alpha = 0, \quad (4)$$

and work it out for a perfect fluid in the form of a gas of relativistic particles in a space-time described by the metric of a spatially flat Friedmann-Lemaître-Robertson-Walker (FLRW) universe. This should allow you to derive how the energy density of the gas scales with FLRW scale factor $a(t)$.

3. Nuclear photodisintegration

The energy per nucleon ϵ_{GDR} necessary to induce a Giant Dipole Resonance (GDR) scales with the nucleus mass number $A > 1$ as

$$\epsilon_{\text{GDR}} \simeq 80A^{-1/3} \quad [\text{MeV}]. \quad (5)$$

Start from the photon energy per nucleon in the nucleus rest frame (NRF) above and calculate its wavelength in the laboratory frame, where the nucleus is an UHECR, say $E_{\text{UHECR}} = 10^{19}$ eV. For nuclei like He, C, N, O, in which part of the electromagnetic spectrum do the corresponding laboratory wavelengths fall?

4. Geometric reconstruction of EAS with a surface array - I:

When entering the atmosphere, UHECRs collide with its atoms producing a cascade of particles called an extensive air shower (EAS) that can reach the ground and trigger an array of surface detectors. The cascade shower front can be approximated by (at least for points not too far away from the shower axis) by a plane propagating in the atmosphere at the speed of light c towards the ground (see figure 1).

- (a) Suppose the shower develops along the direction given by the angles θ and ϕ in spherical coordinates. If it hits the ground and triggers a set of N detectors deployed at positions $\{(x_1, y_1), \dots, (x_N, y_N)\}$, show that the time at which the i -th detector triggers is given by:

$$t_i = T_0 - \left[\frac{u(x_i - x_c) + v(y_i - y_c)}{c} \right], \quad (6)$$

where T_0 is the time at which the shower core (its central part) hits the ground at impact point given by (x_c, y_c) . The variables u and v are called the direction cosines and are given by

$$u = \sin \theta \cos \phi \quad \text{and} \quad v = \sin \theta \sin \phi \quad (7)$$

- (b) In a real experiment, due to measurement uncertainties, the true trigger times t_i will be measured as t_i^m with uncertainties σ_i . Therefore, the values of u and v and, in turn, the arrival direction of the UHECR can be estimated by minimizing a χ^2 :

$$\chi^2 = \sum_{i=1}^N \left(\frac{t_i^m - t_i}{\sigma_i} \right)^2. \quad (8)$$

The minimum of the χ^2 should correspond to a point in the parameter space (T_0, u, v) such that $\frac{\partial \chi^2}{\partial T_0} = 0$, $\frac{\partial \chi^2}{\partial u} = 0$ e $\frac{\partial \chi^2}{\partial v} = 0$. Therefore, show that the solution for the minimum can be put into the form of the linear system (with $w_i = 1/\sigma_i^2$):

$$\underbrace{\begin{pmatrix} \sum_i w_i & \sum_i w_i \bar{x}_i & \sum_i w_i \bar{y}_i \\ \sum_i w_i \bar{x}_i & \sum_i w_i \bar{x}_i^2 & \sum_i w_i \bar{x}_i \bar{y}_i \\ \sum_i w_i \bar{y}_i & \sum_i w_i \bar{x}_i \bar{y}_i & \sum_i w_i \bar{y}_i^2 \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} T_0 \\ u' \\ v' \end{pmatrix}}_{\mathbf{x}} = \underbrace{\begin{pmatrix} \sum_i w_i t_i^m \\ \sum_i w_i \bar{x}_i t_i^m \\ \sum_i w_i \bar{y}_i t_i^m \end{pmatrix}}_{\mathbf{b}}, \quad (9)$$

where $u' = -u/c$, $v' = -v/c$, $\bar{x}_i = x_i - x_c$, $\bar{y}_i = y_i - y_c$ and the sum runs over all triggered detectors.

5. *Geometric reconstruction of EAS with a surface array - II:*

- (a) Write a program in your language of choice (C, C++, Python, Mathematica, Maple, Octave, ROOT, etc) to solve the previous matrix equation. The ASCII files `chuveiros_bari_sig_xxstats.txt` available here:

https://github.com/emourasantos/synthetic_showers.git

where `xx` represents the number of triggered detectors) contain measurements of showers (events) by a hypothetical ground array of detectors, whose resolution in the trigger time is $\sigma_i = 25$ ns. The format of each file line is:

$N \quad \sigma_i(\text{ns}) \quad x_b(\text{m}) \quad y_b(\text{m}) \quad \theta_v(\text{deg}) \quad \phi_v(\text{deg}) \quad x_1(\text{m}) \quad y_1(\text{m}) \quad t_1^m(\text{ns}) \cdots x_N(\text{m}) \quad y_N(\text{m}) \quad t_N^m$

where (θ_v, ϕ_v) are the true values of the shower arrival direction, (x_b, y_b) is an estimate of the shower core impact point at ground calculated as the barycenter of the triggered detectors (using the signal S_i on each detector as a weight)

$$x_b = \frac{\sum_{i=1}^N S_i x_i}{\sum_{i=1}^N S_i}, \quad y_b = \frac{\sum_{i=1}^N S_i y_i}{\sum_{i=1}^N S_i}. \quad (10)$$

Use (x_b, y_b) as an estimate for (x_c, y_c) in the previous equations.

- (b) Each matrix inversion should provide values of u and v at an event-by-event basis. Therefore, you should be able to also get at an event-by-event basis the opening angle $\eta = \arccos(\hat{\mathbf{n}}_v \cdot \hat{\mathbf{n}}_m)$ between the true $\hat{\mathbf{n}}_v$ and reconstructed direction $\hat{\mathbf{n}}_m$. Make histograms of this angle (in deg.) for all the events in `chuveiros_xx_stats.dat`, independent of the number of detectors triggered (there should be a total of 10000 events in the files altogether). This histogram should be well described by a function

$$P(\eta) = C \sin \eta e^{-\frac{\eta^2}{2\sigma_r^2}} \quad (11)$$

where C is a normalization constant and σ_r can be taken as an estimate of the angular resolution of the surface array. Determine the global angular resolution for the array that measured the events in `showers_xx_stats.dat`.

- (c) Make a study of the angular resolution as a function of the number of triggered detectors $N = 4, 5, 6, 7$ and $N \geq 8$. What can you conclude from this study?
- (d) For the complete set of showers, make a histogram of the χ^2 probability

$$P(\geq \chi^2) = \int_{\chi^2}^{\infty} p_n(x) dx, \quad (12)$$

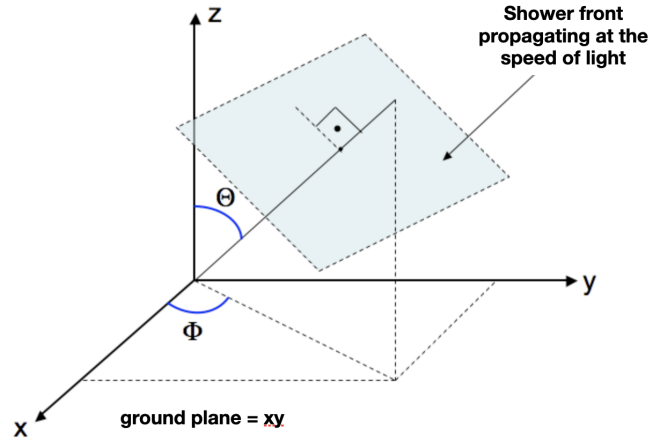


Figure 1: Planar shower front approaching the ground.

where $p_n(x)$ is the pdf of a random χ^2 variable with n degrees of freedom. Be careful to the fact that events have different degrees of freedom. Also prove that $P(\geq \chi^2)$ should, by construction, be a uniform probability distribution. Is your histogram consistent with this property? If not, can you come up with a good explanation for the non-uniformity?